



Further Together

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Table of Contents:

1. Definitions.....	4
2. Abbreviations used:.....	4
3. Introduction.....	5
4. Purpose.....	5
5. Scope	5
6. Background and core functions of Press Trust Malawi.....	5
7. Overall Description	6
8. Assumptions and dependencies.....	8
9. Functional Requirements.....	8
10. Non-core functional requirements.....	9
11. Program and configuration management	10
12. External Interface Requirements.....	43
13. Data Requirements.....	44
14. Non-Functional Requirements.....	45
15. Verification, Acceptance, and Testing.....	49
16. Deliverables expected from the vendor.....	49
17. Appendix	50
18. Process Flow Diagram	76
19. Ownership, Intellectual Property & Sustainability	80
20. Recommended Implementation Timeline	81
21. Milestone-Based Payment Plan	81
22. Document Approval	82

1. Definitions

Cloud Computing	Cloud computing is access to computer system resources, data storage and computing power on demand and without direct management by the user.
Data\Information	This is Information belonging to Press Trust, employee private information and Third-Party information.
Data Centre	This is a dedicated space used to house computer, telecommunications, and data storage systems.
IT Assets	These refer to computing and other devices able to connect to a computer network, Software applications, IT and Information services accessible by devices and applications (e.g., Network infrastructure and internet connectivity)

2. Abbreviations used:

SMS	Scholarship Management System
RBAC	Role Based Access Control
SSL/TLS	Secure Sockets Layer / Transport Layer Security
RPO/RTO	Recovery Point Objective / Recovery Time Objective
MFA	Multi-Factor Authentication
SLA	Service Level Agreement
API	Application Programming Interface

3. Introduction

Press Trust Malawi is a charitable trust established to promote social and economic development in Malawi, with a strong focus on supporting access to education through scholarships and related educational support programmes. The Trust operates as an independent, non-political institution committed to equity, transparency, accountability, and national development impact.

The Scholarship Management System (SMS) is intended to digitise and streamline Press Trust's scholarship administration processes. The system will provide a single, authoritative digital platform to manage the full scholarship lifecycle, from application intake, administration, disbursement oversight, beneficiary monitoring, and reporting. By digitising these processes, Press Trust aims to improve service delivery, reduce manual processing risks, strengthen governance and auditability, and enhance data-driven decision-making.

4. Purpose

The purpose of this Software Requirements Specification (SRS) is to define a complete, clear, and unambiguous set of functional and non-functional requirements for the Scholarship Management System (SMS). This document serves as the authoritative baseline for system design, development, testing, deployment, and acceptance. It is intended to enable software vendors and developers to deliver the SMS with minimal clarification cycles, thereby reducing implementation risk, cost overruns, and project delays.

5. Scope

The Scholarship Management System (SMS) will support Press Trust's scholarship operations from program setup through application intake, approvals, disbursement tracking, beneficiary monitoring, and reporting. The system will provide portals for applicants, external stakeholders, and internal workflows for Press Trust operations, finance, M&E, management, and administrators.

6. Background and core functions of Press Trust Malawi

Press Trust Malawi is a charitable trust with a national mandate to provide funding for social and economic development in Malawi, including education support and other national benefit purposes. Press Trust's operational core functions include:

- Scholarship management: administering merit- and/or needs-based education support, including selection, payments, and beneficiary tracking.
- Stakeholder management: working with schools/institutions, guardians, donors/sponsors, and implementing partners.
- Monitoring, evaluation, and learning: tracking performance, outcomes, and impact metrics for funded beneficiaries and programs.
- Financial stewardship and reporting: ensuring traceable approvals and payments, compliance, and timely reporting to management, board, and stakeholders.

7. Overall Description

7.1 Product perspective

The Scholarship Management System (SMS) must be a web-based information system (with optional mobile-first design) that centralizes scholarship data, workflows, documents, decisions, and financial tracking. It will replace the manual and spreadsheet-based processes and provides controlled access to internal and external users.

7.2 Product functions

At a high level, the SMS shall provide the following capabilities:

- Program configuration and publishing (program setup, forms, rules, timelines, templates).
- Applicant portal (registration, application, document upload, tracking, notifications).
- Eligibility screening and validation (automated and manual overrides with audit trail).
- Entitlements allocation and Inventory management.
- Award management (letters/agreements, acceptance, renewals, lifecycle status).
- Disbursement and financial tracking (schedules, approvals, payment evidence, reconciliation, exceptions).
- Beneficiary monitoring and case management (performance, site visits, interventions, outcomes).
- Reporting and analytics (dashboards, exports, custom reports, dynamic reports).
- Administration and security (RBAC, audit logging, configuration without code, data protection).

7.3 User classes and characteristics

The SMS must support the following user classes. Exact roles can be configured through RBAC.

User Class	Typical Responsibilities	Access Level
Applicant	Create/submit applications; upload documents; track status; accept awards	Internal and External, restricted to own data
Program Officer (Operations)	Configure programs; manage intake; coordinate evaluation; generate letters; track lifecycle	Internal, program-scoped
M&E Officer	Track beneficiary progress, visits, interventions; produce outcome reports	Internal, program/beneficiary-scoped
Finance (Maker/Checker/Approver)	Prepare, approve, and record disbursements;	Internal, maker-checker controls

	attach evidence; reconcile	
Sponsor/Donor	View program portfolio and reports; download approved reporting packs	External/internal; sponsor-scoped
System Administrator	User management; configuration; security policies; reference data	Internal, full admin
Auditor (Read-only)	Review logs, decisions, and payment evidence	Internal/external, read-only with export controls

7.4 Operating environment

The SMS must operate as a secure web application accessible over modern browsers (Chrome/Edge/Firefox) and optimized for mobile devices. The solution may be deployed on cloud infrastructure or on premises, subject to procurement decisions. The system shall support low-bandwidth usage through optimized pages and resumable uploads.

The system shall support deployment on industry-standard operating systems, including low-cost, open-source platforms, with full vendor or enterprise support.

7.5 Hardware

The solution must be capable of operating on either virtualized environments (including on-premise or cloud-based virtual machines) or bare-metal servers. The proposed architecture should be scalable to accommodate future growth in users, data volumes, and transaction loads without significant hardware reconfiguration.

7.6 Software / Operating System

The solution must support deployment on industry-standard operating systems. Use of low-cost, open-source platforms is acceptable, and licensing should not restrict flexibility in operating system choice.

7.7 Database

The solution must support a robust, secure, and proven relational database management system (RDBMS). Acceptable databases may include MySQL, PostgreSQL, Oracle Database, or Microsoft SQL Server, or equivalent enterprise-grade databases. The database should support high availability, data integrity, backup and recovery, and scalability.

7.8 Application Software

The system must be web-based, accessible through standard web browsers without the need for specialized client installations. The application shall utilize open data exchange and interoperability standards (such as RESTful APIs, XML, or JSON) to facilitate integration with third-party systems and future platforms.

8. Assumptions and dependencies

- Press Trust will provide or approve official program policies for configuration.
- Integration with SAGE Evolution, Email Gateway.
- The solution must support hosting at Press Trust offices or any other approved hosting environment/site, without technical or licensing limitations.
- User onboarding will include training and agreed support processes.
- Where historical records are available and of acceptable quality, the proposed solution shall support the migration of up to five (5) years of data from existing Excel-based records into the new system.

9. Functional Requirements

This section provides detailed functional requirements organized by capability area.

9.1 Core Functional Requirements

9.1.1 Applicant and Beneficiary Management

The system shall support the registration and management of applicants and beneficiaries throughout the Scholarship lifecycle. This module shall include:

- Beneficiary biodata and demographic information
- Academic and institutional details
- Linking beneficiaries to schools and academic periods
- Maintenance of schools and institutions master data, including school names, types, districts, locations, contact details, and bank/payment details
- Status management (applicant, active beneficiary, suspended, completed, terminated)
- Secure storage of supporting documentation

9.1.2 Application and Eligibility Management

The system shall support the end-to-end management of Scholarship applications, including:

- Configurable application forms for new and renewal applications
- Application tracking and status updates

9.1.3 Award and Disbursement Management

The system shall support the management of Scholarship awards:

- Award creation, amounts, duration, and conditions
- Payments made directly to schools (tuition, boarding, exam fees, school shoes, school uniform, books and other charges)
- Disbursement schedules, payment tracking, and reconciliation
- Support management, including:
 - Recording of items purchased by the Press Trust

- Issuance of items to beneficiaries
 - Valuation of issued items
 - Capture and storage of beneficiary acknowledgement and signatures confirming receipt
- Linkage of awards and disbursements to funding sources or sponsors.

9.1.4 Monitoring, Reporting, and Audit

The system shall support governance, oversight, and accountability through:

- Monitoring academic progress and compliance with Scholarship conditions.
- Tracking of sponsor-funded and Trust-funded beneficiaries.
- Standard and ad-hoc reporting by beneficiary, school, district, sponsor, and period.
- Budget utilization and commitment reporting.
- Comprehensive system audit logs and reporting.
- The system must capture academic progress and milestone reports, to assess continued eligibility.
- The system must trigger alerts for non-compliance or poor performance, to enable early intervention.
- The system must support continuation, suspension, or termination decisions, to protect funding integrity.

10. Non-core functional requirements

10.1.1 Document & Records Management

- The system must provide secure centralized document storage (to preserve institutional records).
- The system must support version control and access permissions (to prevent unauthorized changes).
- The system must enable advanced search and retrieval (to improve operational efficiency).

10.1.2 Communication & Notifications

- The system must send automated email notifications (To keep applicants and staff informed).
- The system must support bulk communications and templates (To manage large applicant volumes efficiently).
- The system must log all communication activities (To ensure traceability).

10.1.3 Reporting & Analytics

- The system must generate real-time operational dashboards (to support management oversight)
- The system must produce donor and compliance reports (to meet funding requirements).
- The system must support data export in standard formats (PDF, Excel, CSV) - (to facilitate analysis and audits).

10.1.4 System Configuration & Administration

- The system must allow configuration of workflows, forms, and notifications without coding (to ensure flexibility and sustainability).
- The system must support backup and disaster recovery mechanisms (to ensure business continuity).
- The system must provide system performance monitoring tools (to maintain uptime and reliability).

10.1.5 User & Access Management

- The system must support role-based access control (RBAC) – (to ensure users only access functions relevant to their responsibilities).
- The system must allow secure user registration and authentication (To protect applicant and institutional data).
- The system must maintain full audit logs of user activities (to support accountability, investigations, and compliance audits).
- The system must enforce password and security policies (to reduce unauthorized access risks).

11. Program and configuration management

11.1.1 Program Management

The system shall allow Press Trust to create and manage multiple programs (scholarships, bursaries) with flexible configuration so that changes can be made by administrators without vendor code changes.

Req ID	Requirement	Priority	Verification
FR-01	The system shall allow authorized administrators to create a new Scholarship program.	Must	System demo
FR-02	The system shall allow authorized administrators to view and list all Scholarship programs.	Must	System demo
FR-03	The system shall allow authorized administrators to update program details.	Must	System demo
FR-04	The system shall allow authorized administrators to deactivate or archive a Scholarship program.	Must	System demo

FR-05	The system shall allow administrators to configure application opening date per program.	Must	System demo
FR-06	The system shall allow administrators to configure application closing date per program.	Must	System demo
FR-07	The system shall automatically close application submission when the closing date is reached.	Must	System demo
FR-08	The system shall allow administrators to configure required onboarding documents per program.	Must	System demo
FR-09	The system shall allow administrators to configure funding source allocation per program.	Must	System demo
FR-10	The system shall allow administrators to change program status (Draft / Open / Closed / Archived).	Must	System demo
FR-11	The system shall maintain an audit trail of all program configuration changes (who, what, when, before/after).	Should	System demo

- Administrators shall configure program-specific eligibility rules, required documents, and application form fields using a form builder with conditional logic.
- Administrators shall configure evaluation rubrics with weights, minimum thresholds, and tie-break logic.
- Administrators shall configure workflow stages and approvers per program (single-stage, multi-stage, panel, moderation, approval).
- Administrators shall configure templates for communications and documents (emails, award letters, contracts).

11.1.2 User roles, access control, and workflow

The system shall enforce robust role-based access control (RBAC) and workflow assignment to ensure segregation of duties, confidentiality of applications, and auditability of decisions.

Req ID	Requirement	Priority	Verification
FR-12	The system shall allow System Administrator to create user roles.	Must	System demo
FR-13	The system shall allow System Administrator to edit user roles.	Must	System demo
FR-14	The system shall allow System Administrator to deactivate user roles.	Should	System demo
FR-15	The system shall allow System Administrator to assign permissions to a role.	Must	System demo
FR-16	The system shall allow System Administrator to create internal user accounts.	Must	System demo
FR-17	The system shall allow System Administrator to update user account details.	Must	System demo
FR-18	The system shall allow System Administrator to deactivate or suspend user accounts.	Must	System demo
FR-19	The system shall allow System Administrator to assign one or more roles to a user.	Must	System demo
FR-20	The system shall restrict system functionality based	Must	System demo

	on assigned role permissions.		
FR-21	The system shall restrict access to sensitive financial data to authorized finance roles only.	Must	System demo
FR-22	The system shall restrict access to bank account details through masking by default.	Must	System demo
FR-23	The system shall restrict modification of approved disbursement records.	Must	System demo
FR-24	The system shall enforce segregation between disbursement preparation (Maker) and disbursement approval (Checker).	Must	System demo
FR-25	The system shall prevent a user from approving a disbursement they created.	Must	System demo
FR-26	The system shall enforce separation between onboarding approval and financial disbursement approval.	Must	System demo
FR-27	The system shall restrict user access to data based on assigned program(s).	Must	System demo
FR-28	The system shall allow read-only access for internal auditors.	Must	System demo

- The system shall enforce least-privilege access by role, program, and assignment (e.g., reviewers only see assigned applications).
- The system shall enforce segregation of duties between evaluation, approval, and payment processing (maker-checker-approver).
- The system shall provide reviewer conflict-of-interest declarations and block or escalate affected assignments.
- The system shall ensure that user logins are securely managed and that users can only view or access information they are authorized to see.

11.1.3 Applicant intake and submission

The system shall provide an applicant-facing portal to support online submissions, drafts, validation, and automated notifications.

Req ID	Requirement	Priority	Verification
FR-29	The system shall allow authorized users to upload a beneficiary list in CSV format.	Must	System demo
FR-30	The system shall allow users to download a standardized CSV import template.	Must	System demo
FR-31	The system shall allow configurable mapping between CSV columns and system fields during import.	Must	System demo
FR-32	The system shall validate required fields during CSV import.	Must	System demo
FR-33	The system shall validate data types during CSV import (dates, numeric fields, formats).	Must	System demo
FR-34	The system shall validate duplicate beneficiary records during import.	Must	System demo
FR-35	The system shall validate referential data (e.g., school, district, program must exist).	Must	System demo

FR-36	Save-as-draft, autosave, and ability to edit until final submission.	Must	System demo
FR-37	The system shall generate a row-level error log for failed import records.	Must	System demo
FR-38	The system shall generate an import summary report (total uploaded, accepted, rejected).	Must	System demo
FR-39	The system shall allow import of valid records without being blocked by rejected records.	Must	System demo
FR-40	The system shall allow authorized users to view imported beneficiary records.	Must	System demo
FR-41	The system shall allow authorized users to edit imported beneficiary records before onboarding.	Must	System demo
FR-42	The system shall allow authorized users to delete incorrectly imported beneficiary records prior to onboarding.	Must	System demo
FR-43	The system shall allow authorized users to convert an imported beneficiary record into an active Scholarship profile.	Must	System demo
FR-44	The system shall enforce completion of mandatory onboarding fields before activation.	Must	System demo
FR-45	The system shall allow uploading of onboarding supporting documents.	Must	System demo

FR-46	The system shall assign a unique internal beneficiary identifier upon onboarding.	Must	System demo
FR-47	The system shall allow authorized users to activate a beneficiary record.	Must	System demo
FR-48	The system shall allow authorized users to suspend a beneficiary record with mandatory reason capture.	Must	System demo
FR-49	The system shall allow authorized users to reinstate a suspended beneficiary record.	Must	System demo
FR-50	The system shall allow authorized users to terminate or close a beneficiary record with mandatory reason capture.	Must	System demo
FR-51	The system shall prevent financial disbursement to beneficiaries whose status is not Active.	Must	System demo
FR-52	The system shall maintain an audit trail for all beneficiary data changes.	Must	System demo
FR-53	The system shall maintain an audit trail for all status changes.	Must	System demo
FR-54	The system shall prevent duplicate onboarding of the same beneficiary based on defined uniqueness criteria (e.g., National ID or Examination ID).	Must	System demo
FR-55	Automated submission confirmation (email) with reference	Must	System demo

	number and next steps.		
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- Applicants shall create accounts using email and/or phone number; system shall support password reset and Multi-Factor Authentication (MFA).
- Applications shall support save-as-draft, autosave, and resume later, including partial document uploads.
- The system shall validate data formats and completeness and prevent submission until mandatory requirements are met.
- The system shall support confirmations and status updates via email/in-app notifications.

11.1.4 Data management

The system shall maintain complete and accurate master and transactional data for applicants, guardians, schools, sponsors, programs, applications, awards, payments, and beneficiary performance, with full audit trails.

Req ID	Requirement	Priority	Verification
FR-56	The system shall allow authorized users to create beneficiary records.	Must	System demo
FR-57	The system shall allow authorized users to view beneficiary records.	Must	System demo
FR-58	The system shall allow authorized users to update beneficiary records.	Must	System demo
FR-59	The system shall allow authorized users to deactivate or archive beneficiary records.	Must	System demo
FR-60	The system shall enforce uniqueness rules to prevent duplicate beneficiaries (e.g., National ID + Exams ID + Intake).	Must	System demo
FR-61	The system shall allow authorized users to create guardian/next-of-kin details linked to a beneficiary.	Must	System demo
FR-62	The system shall allow authorized users to update guardian/next-	Must	System demo

	of-kin details linked to a beneficiary.		
FR-63	The system shall allow authorized users to view guardian/next-of-kin details linked to a beneficiary.	Must	System demo
FR-64	The system shall allow authorized users to deactivate guardian/next-of-kin records.	Must	System demo
FR-65	The system shall allow authorized users to create school/institution records.	Must	System demo
FR-66	The system shall allow authorized users to view school/institution records.	Must	System demo
FR-67	The system shall allow authorized users to update school/institution records.	Must	System demo
FR-68	The system shall allow authorized users to deactivate or archive school/institution records.	Must	System demo
FR-69	The system shall allow authorized users to create a bank account record linked to a school.	Must	System demo
FR-70	The system shall allow authorized users to view bank account records linked to a school.	Must	System demo
FR-71	The system shall allow authorized users to update bank account records linked to a school.	Must	System demo
FR-72	The system shall allow authorized users to deactivate bank account records linked to a school.	Must	System demo
FR-73	The system shall require approval (maker-checker) for changes to active school bank account details.	Must	System demo

FR-74	The system shall allow authorized users to create reference data records (e.g., districts, academic year/term, item categories).	Must	System demo
FR-75	The system shall allow authorized users to update reference data records.	Must	System demo
FR-76	The system shall allow authorized users to view reference data records.	Must	System demo
FR-77	The system shall allow authorized users to deactivate reference data records.	Must	System demo
FR-78	The system shall validate mandatory fields for beneficiary records.	Must	System demo
FR-79	The system shall validate field formats for key data elements (e.g., phone, dates, numeric fields).	Must	System demo
FR-80	The system shall enforce referential integrity between related records (e.g., beneficiary must reference a valid school and program).	Must	System demo
FR-81	The system shall prevent deletion of records that are referenced by transactions (e.g., disbursements).	Must	System demo
FR-82	The system shall mask bank account details by default.	Must	System demo
FR-83	The system shall restrict viewing of unmasked bank account details to authorized roles only.	Must	System demo
FR-84	The system shall log every access to unmasked bank account details.	Must	System demo
FR-85	The system shall log creation of beneficiary records (who, when).	Must	System demo

FR-86	The system shall log updates to beneficiary records including before/after values.	Must	System demo
FR-87	The system shall log deactivation or archiving of beneficiary records.	Must	System demo
FR-88	The system shall allow authorized users to view audit logs for a selected record.	Must	System demo
FR-89	The system shall allow authorized users to export audit logs for audit purposes.	Must	System demo
FR-90	Capture and store detailed student/applicant information (demographics, contacts, location, identifiers).	Must	System demo
FR-91	Capture and store parent/guardian information, including relationship, contacts, and consent where required.	Must	System demo
FR-92	Capture and manage school/institution profiles (contacts, location, accreditation/registration status, identifiers).	Must	System demo
FR-93	Capture and manage scholarship sponsor/donor profiles (funding rules, restrictions, reporting needs).	Must	System demo
FR-94	Capture and securely store banking details for schools and parents/guardians (with access controls and masking).	Must	Demo & security review
FR-95	Maintain student performance indicators over time (term/semester results, attendance, progression, completion).	Must	System demo

- The system shall maintain master data for districts, schools, programs, sponsors, and support categories (fees, books, transport, etc.).
- Sensitive fields (e.g., bank details) shall be masked by default and accessible only to authorized finance roles and passwords shall be required to open the sensitive fields.
- The system shall maintain a complete audit trail for create/update/delete actions on key records (who, what, when, before/after).

11.1.5 Document management

The system shall support secure upload, storage, verification, and controlled access to supporting documents.

Req ID	Requirement	Priority	Verification
FR-96	The system shall allow authorized users to upload documents linked to a beneficiary record.	Must	System demo
FR-97	The system shall allow authorized users to upload documents linked to a disbursement record.	Must	System demo
FR-98	The system shall virus-scan documents upon upload.	Must	System demo
FR-99	The system shall store uploaded documents with metadata (document type, owner record, upload date, uploaded by).	Should	System demo
FR-100	The system shall enforce configurable file size limits.	Must	System demo
FR-101	The system shall allow authorized users to view document metadata.	Must	System demo
FR-102	The system shall allow authorized	Must	System demo

	users to download documents.		
FR-103	The system shall restrict document access based on user role permissions.	Must	System demo
FR-104	The system shall log all document download activities.	Must	System demo
FR-105	The system shall allow authorized users to set document status to "Pending".	Must	System demo
FR-106	The system shall allow authorized users to set document status to "Verified".	Must	System demo
FR-107	The system shall allow authorized users to set document status to "Rejected".	Must	System demo
FR-108	The system shall require capture of a rejection reason when a document is marked as Rejected.	Must	System demo
FR-109	The system shall allow authorized users to upload a new version of an existing document.	Must	System demo
FR-110	The system shall retain previous document versions for audit purposes.	Must	System demo
FR-111	The system shall display document version history.	Must	System demo
FR-112	The system shall allow authorized users to define expiry dates for documents.	Must	System demo

FR-113	The system shall automatically flag expired documents.	Must	System demo
FR-114	The system shall allow authorized users to request re-upload of expired or rejected documents.	Must	System demo
FR-115	The system shall prevent deletion of documents that are linked to approved financial transactions.	Must	System demo
FR-116	The system shall allow only authorized roles to delete documents prior to approval.	Must	System demo
FR-117	The system shall maintain an immutable audit trail of all document-related actions (upload, update, delete, verification, download).	Must	System demo
FR-118	The system shall support bulk document upload linked to a batch of beneficiaries.	Must	System demo
FR-119	The system shall prevent disbursement processing if mandatory supporting documents are missing or not verified.	Must	System demo

- Documents shall be virus-scanned on upload and stored with metadata (type, owner, date, expiry, version).
- The system shall provide a verification workflow (Pending/Verified/Rejected) with reason capture and re-upload requests.

- Downloads/exports shall be controlled and logged; bulk downloads shall be restricted to specific roles.

11.1.6 Eligibility screening, review, and evaluation

The system shall support automated and manual eligibility screening, reviewer scoring, ranking/shortlisting, and committee approvals.

Req ID	Requirement	Priority	Verification
FR-120	The system shall validate completeness of mandatory beneficiary data fields before onboarding.	Must	System demo
FR-121	The system shall validate beneficiary uniqueness against existing records in the system.	Must	System demo
FR-122	The system shall validate that referenced schools exist in the master data.	Must	System demo
FR-123	The system shall flag inconsistencies between imported data and existing master data.	Must	System demo
FR-124	The system shall allow authorized users to correct validation errors prior to onboarding.	Must	System demo
FR-125	The system shall allow authorized users to confirm that a beneficiary record has passed internal validation checks.	Must	System demo
FR-126	The system shall allow authorized users to approve a beneficiary record for onboarding.	Must	System demo
FR-127	The system shall prevent	Must	System demo

	onboarding of beneficiary records that are not internally approved.		
FR-128	The system shall allow authorized users to flag a beneficiary record as "Exception".	Must	System demo
FR-129	The system shall require capture of a reason when marking a beneficiary record as Exception.	Must	System demo
FR-130	The system shall allow authorized users to resolve flagged exceptions.	Must	System demo
FR-131	The system shall maintain an audit trail of exception status changes.	Must	System demo
FR-132	The system shall allow segregation between validation confirmation and onboarding approval roles.	Must	System demo
FR-133	The system shall prevent a user from approving onboarding for a beneficiary record they validated.	Must	System demo
FR-134	The system shall maintain an audit trail of all validation and approval actions.	Must	System demo
FR-135	The system shall allow authorized users to generate a validated beneficiary register report prior to disbursement processing.	Must	System demo

- Eligibility engine shall produce clear reasons for ineligibility and allow authorized override with justification.
- Reviewer screens shall support criterion-level scoring, comments, and attachments/notes.
- The system shall produce ranked lists and shortlists, including tie-handling and moderation rules.
- Committee decisions shall capture outcomes (Approved/Waitlisted/Rejected) with reasons and decision minutes (optional).

11.1.7 Award and contract management

The system shall generate award letters/contracts, manage acceptance and consent workflows, and track the award lifecycle including renewals.

Req ID	Requirement	Priority	Verification
FR-136	The system shall allow authorized users to create a Scholarship award record linked to an approved beneficiary.	Should	System demo
FR-137	The system shall allow authorized users to view Scholarship award records.	Should	System demo
FR-138	The system shall allow authorized users to update Scholarship award details prior to financial activation.	Must	System demo
FR-139	The system shall prevent modification of Scholarship award amount after first disbursement unless approved by authorized role.	Must	System demo
FR-140	The system shall generate Scholarship award confirmation letters using configurable templates.	Must	System demo
FR-141	The system shall store generated award letters in the beneficiary record.	Must	System demo
FR-142	The system shall allow authorized users to re-generate award letters where required.	Must	System demo
FR-143	The system shall allow authorized	Must	System demo

	users to activate a Scholarship award.		
FR-144	The system shall prevent financial disbursement before award activation.	Must	System demo
FR-145	The system shall allow authorized users to suspend a Scholarship award with mandatory reason capture.	Must	System demo
FR-146	The system shall allow authorized users to reinstate a suspended Scholarship award.	Must	System demo
FR-147	The system shall allow authorized users to close or terminate a Scholarship award with mandatory reason capture.	Must	System demo
FR-148	The system shall allow authorized users to define award start date.	Must	System demo
FR-149	The system shall allow authorized users to define award end date.	Must	System demo
FR-150	The system shall automatically flag awards reaching end date.	Must	System demo
FR-151	The system shall allow authorized users to renew an award for a new period.	Must	System demo
FR-152	The system shall maintain linkage between original award and renewed award records.	Must	System demo

FR-153	The system shall require linkage of each award to a funding source or budget line.	Must	System demo
FR-154	The system shall prevent award creation if funding allocation is insufficient.	Must	System demo
FR-155	The system shall update program budget balance upon award approval.	Must	System demo
FR-156	The system shall maintain an audit trail of all award record changes.	Must	System demo
FR-157	The system shall log all award status changes.	Must	System demo

- Award letters/contracts shall be generated from templates with variable substitution and stored in the record.
- Acceptance workflow shall capture applicant acceptance plus guardian/school acknowledgements where applicable.
- Renewals shall be configurable and triggered by performance/compliance milestones with reminders and reassessment workflow.
- Awards shall have lifecycle statuses (Draft, Submitted, Under Review, Approved, Awarded, Active, Suspended, Completed, Closed).

11.1.8 Financial tracking and disbursement

The system shall capture disbursement schedules, multi-level payment approvals, and payment evidence, supporting multiple payee models and reconciliation.

Req ID	Requirement	Priority	Verification
FR-158	The system shall allow authorized users to create a disbursement request linked to an active Scholarship award.	Must	System demo
FR-159	The system shall prevent creation of a disbursement	Must	System demo

	request for inactive beneficiaries.		
FR-160	The system shall allow authorized users to specify disbursement amount.	Must	System demo
FR-161	The system shall allow authorized users to specify disbursement category (fees, uniform, books)	Must	System demo
FR-162	The system shall allow authorized users to specify payee type (school, guardian, vendor).	Must	System demo
FR-163	The system shall prevent disbursement amounts from exceeding approved award balance.	Must	System demo
FR-164	The system shall update remaining award balance after each approved disbursement.	Must	System demo
FR-165	The system shall prevent disbursement that exceeds available program budget.	Must	System demo
FR-166	The system shall prevent duplicate disbursement for the same beneficiary, category, and academic period.	Must	System demo
FR-167	The system shall allow override of duplicate prevention only with authorized approval and	Must	System demo

	mandatory justification.		
FR-168	The system shall require disbursement requests to be approved before payment confirmation.	Must	System demo
FR-169	The system shall prevent a user from approving a disbursement they created.	Must	System demo
FR-170	The system shall allow multi-level approval configuration (e.g., Maker-Checker-Approver).	Must	System demo
FR-171	The system shall record approval timestamps and approving user details.	Must	System demo
FR-172	The system shall allow authorized users to update disbursement payment status (Requested / Approved / Paid / Failed / Reconciled).	Must	System demo
FR-173	The system shall prevent status change from Paid to Requested.	Must	System demo
FR-174	The system shall require mandatory reason capture when marking a disbursement as Failed.	Must	System demo
FR-175	The system shall allow authorized users to upload payment evidence	Must	System demo

	linked to a disbursement.		
FR-176	The system shall prevent marking a disbursement as Paid unless payment evidence is attached.	Must	System demo
FR-177	The system shall log upload and modification of payment evidence.	Must	System demo
FR-178	The system shall retrieve payee bank details only from approved master records.	Must	System demo
FR-179	The system shall prevent editing of payee bank details during disbursement processing.	Must	System demo
FR-180	The system shall allow authorized users to record returned funds.	Must	System demo
FR-181	The system shall allow authorized users to record disbursement reversals.	Must	System demo
FR-182	The system shall update award balance automatically upon reversal.	Must	System demo
FR-183	The system shall allow authorized users to mark disbursements as reconciled.	Must	System demo
FR-184	The system shall generate disbursement reconciliation reports.	Must	System demo
FR-185	The system shall prevent	Must	System demo

	modification of reconciled disbursement records.		
FR-186	The system shall maintain a complete audit trail of all disbursement actions.	Must	System demo
FR-187	The system shall log all changes to disbursement amounts.	Must	System demo
FR-188	The system shall log all approval decisions.	Must	System demo

- Disbursement schedules shall support one-off or periodic payments, with tranches linked to milestones or academic terms.
- Payment workflow shall enforce maker-checker-approver roles and capture evidence (vouchers, invoices, bank confirmations).
- The system shall support exceptions (failed payments, returned funds, partial payments) and reconciliation statuses.
- The system shall support payments to schools, guardians, and vendors/suppliers, including split payments if policy allows.

11.1.9 Monitoring & evaluation and case management

The system shall enable ongoing monitoring of beneficiaries, including performance tracking, flags for at-risk beneficiaries, interventions, visits, and outcome metrics.

Req ID	Requirement	Priority	Verification
FR-189	The system shall allow authorized users to record academic performance per beneficiary per academic period and per school.	Must	System demo
FR-190	The system shall allow authorized users to update academic performance records.	Should	System demo
FR-191	The system shall allow authorized	Should	System demo

	users to view historical academic performance records.		
FR-192	The system shall prevent deletion of academic performance records once disbursement has occurred for that period.	Must	System demo
FR-193	The system shall allow authorized users to record attendance status per beneficiary per academic period.	Must	System demo
FR-194	The system shall allow authorized users to record academic progression status (Promoted / Repeated / Completed / Dropped).	Must	System demo
FR-195	The system shall allow authorized users to update progression status with mandatory reason capture where applicable.	Must	System demo
FR-196	The system shall automatically flag beneficiaries whose performance falls below configured threshold.	Must	System demo
FR-197	The system shall allow authorized users to manually flag a beneficiary as At-Risk.	Must	System demo
FR-198	The system shall require reason capture when manually flagging a	Must	System demo

	beneficiary as At-Risk.		
FR-199	The system shall allow authorized users to remove At-Risk flag with justification.	Must	System demo
FR-200	The system shall allow authorized users to record intervention actions linked to a beneficiary.	Must	System demo
FR-201	The system shall allow authorized users to assign responsibility for an intervention action.	Must	System demo
FR-202	The system shall allow authorized users to define due dates for intervention actions.	Must	System demo
FR-203	The system shall allow authorized users to update intervention status (Open / In Progress / Closed).	Must	System demo
FR-204	The system shall prevent deletion of intervention records.	Must	System demo
FR-205	The system shall allow authorized users to record monitoring visits linked to beneficiaries or schools.	Must	System demo
FR-206	The system shall allow authorized users to upload visit reports or supporting documents.	Must	System demo
FR-207	The system shall allow authorized	Must	System demo

	users to record findings from monitoring visits.		
FR-208	The system shall allow authorized users to record follow-up actions from monitoring visits.	Must	System demo
FR-209	The system shall allow authorized users to record completion status of beneficiaries.	Must	System demo
FR-210	The system shall allow authorized users to record graduation outcomes.	Must	System demo
FR-211	The system shall allow authorized users to record exit reasons for beneficiaries who discontinue.	Must	System demo
FR-212	The system shall allow authorized users to generate outcome statistics (completion rate, dropout rate, progression rate).	Must	System demo
FR-213	The system shall prevent renewal of Scholarship awards where performance criteria are not met (if renewal is configured).	Must	System demo
FR-214	The system shall maintain an audit trail of all monitoring and intervention actions.	Must	System demo
FR-215	The system shall allow authorized users to export monitoring and intervention logs.	Must	System demo

- The system shall track term/semester performance, attendance, progression, and completion per beneficiary.
- At-risk rules shall be configurable (e.g., below threshold, non-attendance) and trigger alerts and intervention logs.
- M&E shall capture site visits, findings, attachments, actions, owners, and due dates.
- Outcome metrics shall be program-configurable and reportable (completion rates, dropout rates, graduation, etc.).

11.1.10 Reporting and analytics

The system shall provide dashboards, standard reports, ad-hoc reporting, and exports for operational, financial, and impact reporting.

Req ID	Requirement	Priority	Verification
FR-216	The system shall provide a standard beneficiary register report.	Must	System demo
FR-217	The system shall provide a standard awards report.	must	Technical documentation & demo (if available)
FR-218	The system shall provide a standard disbursements report.	Must	System demo
FR-219	The system shall provide a standard budget utilization report.	Must	System demo
FR-220	The system shall provide a standard school-level payments report.	Must	System demo
FR-221	The system shall provide a standard M&E outcomes report.	Must	System demo
FR-222	The system shall allow users to filter reports by program.	Must	System demo
FR-223	The system shall allow users to filter reports by academic period.	Must	System demo
FR-224	The system shall allow users to filter reports by district.	Must	System demo

FR-225	The system shall allow users to filter reports by school.	Must	System demo
FR-226	The system shall allow users to filter reports by beneficiary status.	Must	System demo
FR-227	The system shall allow authorized users to select data fields for an ad-hoc report.	Must	System demo
FR-228	The system shall allow authorized users to apply filters to an ad-hoc report.	Must	System demo
FR-229	The system shall allow authorized users to run an ad-hoc report on demand.	Must	System demo
FR-230	The system shall allow authorized users to save an ad-hoc report definition as a template.	Must	System demo
FR-231	The system shall allow authorized users to update saved report templates.	Must	System demo
FR-232	The system shall allow authorized users to delete saved report templates.	Must	System demo
FR-233	The system shall allow users to export reports to CSV format.	Must	System demo
FR-234	The system shall allow users to export reports to Excel format.	Must	System demo
FR-235	The system shall allow users to export reports to PDF format.	Must	System demo
FR-236	The system shall include the applied filters in exported report outputs.	Must	System demo
FR-237	The system shall display a management dashboard showing key Scholarship KPIs.	Must	System demo

FR-238	The system shall display total active beneficiaries by program.	Must	System demo
FR-239	The system shall display total disbursements by period.	Must	System demo
FR-240	The system shall display budget utilization by program.	Must	System demo
FR-241	The system shall display at-risk beneficiary counts.	Must	System demo
FR-242	The system shall allow users to drill down from dashboard KPIs to underlying beneficiary lists.	Must	System demo
FR-243	The system shall allow users to drill down from dashboard KPIs to underlying disbursement records.	Must	System demo
FR-244	The system shall restrict access to financial reports to authorized finance roles only.	Must	System demo
FR-245	The system shall restrict access to beneficiary personal data fields based on role permissions.	Must	System demo
FR-246	The system shall log report export actions (who exported, when, report name, format).	Must	System demo
FR-247	The system shall allow auditors to view report export logs.	Must	System demo
FR-248	The system shall allow authorized users to schedule automated report generation.	Must	System demo

FR-249	The system shall allow authorized users to schedule automated report delivery to internal user groups.	Must	System demo
FR-250	The system shall provide a dynamic report builder interface that allows authorized users to create reports by selecting fields using drag-and-drop.	Must	System demo
FR-251	The system shall allow users to drag and drop available data fields into a report layout (rows/columns/filters).	Must	System demo
FR-252	The system shall allow users to add and configure filters within the dynamic report builder.	Must	System demo
FR-253	The system shall allow users to save a dynamic report definition as a reusable template.	Must	System demo
FR-254	The system shall allow users to edit a saved dynamic report template.	Must	System demo
FR-255	The system shall allow users to delete a saved dynamic report template.	Must	System demo
FR-256	The system shall allow users to run a saved dynamic report template on demand.	Must	System demo
FR-257	The system shall allow users to export dynamic report outputs in CSV and Excel formats.	Must	System demo
FR-258	The system shall restrict access to	Must	System demo

	dynamic report builder functionality based on role permissions.		
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- Dashboards shall show pipeline and performance KPIs with drill-down by program, district, gender, school, sponsor, and time period.
- Reports shall be exportable to Excel/PDF/CSV with filter context.
- Ad-hoc reporting shall allow users to select fields, apply filters, save views, and share report definitions subject to RBAC.

11.1.11 Integration and communication

The system shall support configurable notifications and provide APIs and import/export utilities for integration with external systems.

Req ID	Requirement	Priority	Verification
FR-259	The system shall send automated email notifications based on configured events.	Must	System demo
FR-260	The system shall allow administrators to configure notification templates.	Must	System demo
FR-261	The system shall allow administrators to edit notification templates.	Must	System demo
FR-262	The system shall log all notification events (recipient, date, channel, status).	Must	System demo
FR-263	The system shall allow administrators to activate or deactivate notification triggers.	Must	System demo
FR-264	The system shall expose REST APIs for integration with approved financial systems.	Must	System demo
FR-265	The system shall allow retrieval of approved	Must	System demo

	disbursement data through secure API endpoints.		
FR-266	The system shall require authentication and authorization for all API access.	Must	System demo
FR-267	The system shall log all API access activities.	Must	System demo
FR-268	The system shall allow authorized users to upload CSV files for bulk data import.	Must	System demo
FR-269	The system shall allow users to download CSV import templates.	Must	System demo
FR-270	The system shall allow configurable field mapping during CSV import.	Must	System demo
FR-271	The system shall validate imported CSV records.	Must	System demo
FR-272	The system shall generate row-level error logs for failed import records.	Must	System demo
FR-273	The system shall generate an import summary report.	Must	System demo
FR-274	The system shall encrypt SMTP credentials at rest.	Must	System demo
FR-275	The system shall allow administrators to configure SMTP settings.	Must	System demo

- Notifications shall be template-driven and event-based (submission, missing docs, decisions, renewals, payment updates).
- The system shall expose REST APIs (or equivalent) for integrations and provide bulk import/export with mapping and error logs.

12. External Interface Requirements

12.1 User interfaces

The system shall provide secure, role-based internal user interfaces accessible via web browser. The interfaces shall be responsive, optimized for low-bandwidth environments, and support desktop and mobile device access.

12.1.1 Operations Portal

The Operations Portal shall support:

- Program configuration and management
- Government beneficiary list import
- Beneficiary onboarding
- Award creation and activation
- Beneficiary registry management
- Exception handling and validation confirmation

12.1.2 Finance Portal

The Finance Portal shall support:

- Disbursement request creation
- Maker–Checker–Approver workflows
- Payment status tracking
- Payment evidence upload
- Reversal and reconciliation processing
- Financial reporting and exports

12.1.3 Monitoring & Evaluation (M&E) Portal

The M&E Portal shall support:

- Academic performance entry
- Attendance tracking
- At-risk flagging
- Intervention logging
- Site visit recording
- Outcome tracking and dashboards

12.1.4 Administration Portal

The Administration Portal shall support:

- User account management
- Role and permission configuration
- Reference data management
- Template management (letters, notifications)
- System configuration settings
- Audit log access

12.1.5 Management Dashboard

The system shall provide a Management Dashboard showing:

- Active beneficiaries
- Budget utilization
- Disbursement summaries
- At-risk beneficiary counts

- Program performance indicators

12.1.6 Authentication & Security Controls

The system shall:

- Provide secure login functionality for all internal users.
- Provide password reset functionality.
- Enforce multi-factor authentication (MFA) during login.
- Enforce session timeout after configurable inactivity period.
- Prevent access to restricted modules without proper role authorization.

12.2 Hardware interfaces

The SMS shall not require specialized hardware. Optional integrations may include scanners for document digitization and biometric devices only if explicitly requested in future phases.

12.3 System integrations

The SMS should support integration with the following systems where applicable:

- Future integration with Accounting/ERP (e.g., SAGE Evolution): exchange of approved budgets, payment vouchers, and reconciliation references.
- Email service: transactional emails and templated communications.
- Future integration with Payment rails/banking: optional integration for electronic payments and payment status callbacks.
- Identity services: optional single sign-on (SSO) for internal users (e.g., Windows Domain Controller, Azure AD) if available.
- Future integration with financial institutions

12.4 Communications interfaces

The system shall support configurable notification channels (email, in-app). Each notification shall be logged with timestamp, recipient, channel, template used, and delivery status.

13. Data Requirements

13.1 Core entities

Entity	Description / Key Data
Program	Program metadata, budget ceiling, funding source, configuration settings, status.
Beneficiary	Beneficiary biodata, identifiers, contact details, linked program, status (Imported, Active, Suspended, Closed).
School / Institution	School profile, registration status, location, contact details, linked bank accounts.
School Bank Account	Bank name, branch, account number (masked), active status, approval status.

Funding Source	Donor/funding allocation linked to program and awards.
Award	Approved Scholarship allocation linked to beneficiary and funding source, amount, duration, lifecycle status.
Disbursement	Payment request, category, payee, amount, status, approval records, payment evidence.
Disbursement Item	Catalog of allowable support categories (fees, uniforms, books, etc.).
Academic Performance	Term results, attendance status, progression status.
Intervention / Case Log	Recorded actions, assigned officer, due dates, resolution status.
Monitoring Visit	Visit date, findings, attachments, follow-up actions.
Audit Log	Immutable log of user actions (create, update, approve, export, status change).
Notification Log	Record of sent notifications (email), timestamp, recipient, delivery status.
Reference Data	Districts, academic periods, item categories, program types, status codes.

13.2 Data quality and validation

- The system shall enforce field-level validation, including formats for emails, phone numbers, dates, and numeric ranges.
- The system shall enforce uniqueness constraints where applicable (e.g., applicant identifier per program, user email).
- The system shall maintain referential integrity between related entities (e.g., application must reference an existing program).
- The system shall support data completeness indicators (e.g., missing mandatory fields/documents).

13.3 Data retention and archival

Press Trust shall define retention periods. The system shall support configurable retention rules and secure archival of closed records. Records containing minors' data shall follow applicable legal and policy requirements.

14. Non-Functional Requirements

Non-functional requirements define quality attributes and constraints for the SMS.

NFR ID	Requirement	Priority	Verification
NFR-01	Role-based security and access control aligned to least privilege.	Must	Security review & demo
NFR-02	Encryption of data in transit (TLS) and at rest (database and document storage).	Must	Technical documentation

NFR-03	Compliance with applicable data protection/privacy requirements, including consent management for minors where relevant.	Must	Compliance statement & policy review
NFR-04	System availability target (e.g., ≥99.5%) with vendor support SLAs and incident response times.	Should	SLA review
NFR-05	Scalable architecture to support peak application submissions and large document volumes without degradation.	Must	Architecture review & performance evidence
NFR-06	Responsive, mobile-friendly interface suitable for low-bandwidth environments (optimized pages and uploads).	Must	System demo
NFR-07	Audit logs for key transactions and decisions (eligibility, scoring, approvals, payments) and export for audit.	Must	System demo
NFR-08	Automated backups and disaster recovery plan with defined RPO/RTO; tested restoration procedures.	Must	DR documentation
NFR-09	Maintainability/configurability: admin-driven configuration for forms, workflows, rubrics, templates without code changes.	Must	Demo & technical documentation
NFR-10	Vendor support model with defined SLAs, documentation, and training plan for administrators and users.	Should	SLA & support documentation
NFR-11	The system shall support concurrent access by multiple users, allowing simultaneous logins without user or licensing restrictions.	Must	System demo

14.1 Security and privacy

- Authentication shall support strong passwords, configurable lockout after 3 failed attempts, and MFA for all users.
- The system shall ensure that user logins are securely managed and that users can only view or access information they are authorized to see.

- Sessions shall be securely managed and terminated on logout; browser back navigation shall not restore authenticated pages.
- Audit logs shall capture security events and all key business actions; logs shall be tamper-evident and exportable for audit.
- File uploads shall be virus-scanned; file size limits shall be configurable (recommended 50–100MB per file unless policy requires more).
- The system shall ensure that all data exchanged over the network is encrypted using a secure SSL/TLS certificate.
- The system shall support field-level access control to restrict viewing of sensitive fields (e.g., bank account numbers, national IDs) based on user role.
- The system shall log all data export activities, including user, date/time, report name, and file format.
- The system shall enforce segregation of duties between disbursement creation and disbursement approval.
- The system shall support configurable data retention periods and secure archival of inactive records.
- All API endpoints shall require secure authentication and authorization.
- The system shall log all access to sensitive data fields, including user, timestamp, and action performed.
- The system shall enforce approval workflow for changes to stored bank account details.
- The system shall retain audit logs for a configurable period in accordance with organizational policy.
- Audit logs shall be protected from modification or deletion by unauthorized users.
- Access to audit log configuration and management shall be restricted to authorized administrative roles only.
- The system shall generate alerts in the event of audit logging failure or backup failure.
- The system shall synchronize system time with an approved time source to ensure accurate audit timestamps.
- Passwords shall be stored using secure one-way hashing and shall never be stored in plaintext.
- The system shall support secure configuration baselines and removal of default or unused system accounts.
- The system shall support regular security patching of application and underlying components in accordance with defined patch management policy.
- The system shall prevent sensitive data (e.g., full bank account numbers) from being written to system logs.

14.2 Performance, availability, and scalability

- The system shall support peak disbursement, bulk import, and reporting periods without performance degradation.
- Target response times for common operations (view list, open application, save draft) should be ≤ 3 seconds under normal load.
- Availability target should be $\geq 99.5\%$ per month excluding planned maintenance, subject to hosting model.
- The system shall support concurrent access by multiple internal users without performance degradation.
- The system shall support bulk CSV imports of beneficiary records within defined volume thresholds without timeout.
- Standard operational reports shall execute within defined response time thresholds under normal system load.
- The system shall support monitoring and alerting for system availability and critical background jobs.
- Backup processes shall generate automated alerts upon failure.
- Backups shall be automated; disaster recovery shall define RPO/RTO and include documented and periodically tested restoration procedures.

14.3 Usability, accessibility, and localization

- User interfaces shall be mobile-friendly and usable on low-bandwidth networks, including resumable uploads and compressed previews.
- The system shall provide clear, user-friendly error messages that do not expose server internals; technical errors shall be logged for support.
- The system shall support accessibility considerations (readable fonts, keyboard navigation, contrast, screen-reader labels) where feasible.
- The system shall support configurable date/time formats and can be extended to support local languages if required.
- The system shall dynamically render user interfaces based on assigned roles and permissions.
- The system shall provide real-time validation feedback during data entry to reduce data entry errors.
- The system shall provide progress indicators for long-running operations such as bulk imports and report generation.

14.4 Data Governance & Quality

- The system shall enforce consistent validation rules across manual entry, CSV import, and API interfaces.
- The system shall provide duplicate detection mechanisms for beneficiaries and payees.
- The system shall support exception queues for resolution of flagged data inconsistencies.
- The system shall prevent deletion of records that are linked to financial transactions.

14.5 Maintainability & Supportability

- The system shall support separate development, testing, and production environments.
- The system shall allow configuration of forms, workflows, templates, and reports without vendor code changes.
- The system shall provide documented API interfaces where integrations are supported.
- The vendor shall provide defined service level agreements (SLAs) for incident response and resolution.
- The system shall not impose vendor lock-in and shall allow export of core datasets in open formats.

15. Verification, Acceptance, and Testing

Each requirement shall be verified using one or more of the following methods: demonstration, functional testing, document review, security testing, and performance testing. The vendor shall provide a test plan, test cases, and evidence of testing.

The vendor shall support the development, validation, and refinement of detailed business process flows, data flow diagrams, and related system design artefacts based on the agreed functional requirements. This shall include collaboration with the client during implementation and demonstration sessions to ensure that all process flows and data models accurately reflect operational needs and system functionality.

16. Deliverables expected from the vendor

- Solution architecture and hosting design (including scalability, security controls, DR).
- Configured SMS application and source code (as per procurement terms), including admin manuals and user guides.
- API documentation and data import/export templates.
- Deployment and data migration plan.
- Training plan and handover package.
- Support and maintenance SLA proposal (incident response, bug fix timelines, enhancements).

17. Appendix

17.1 Appendix A: Detailed Requirement Catalogue

This appendix provides a more detailed set of requirements derived from the SMS requirements matrix to support API design, detailed testing, and implementation planning. These requirements are not exhaustive and may be refined or supplemented during system demonstrations and solution discussions. Vendors may propose additional requirements or enhancements; however, none of the requirements specified herein shall be removed or excluded.

FR_ID	Module	Feature / Use Case	Requirement (What the system shall do)	Actors / Roles	Priority	Verification	Acceptance Criteria
FR-AUTH-001	Authentication	User Login	The system shall authenticate internal users using email/username and password.	All users	High	Test	Valid credentials succeed; invalid credentials return a generic error without revealing whether the account exists.
FR-AUTH-002	Authentication	Multi-Factor Authentication	The system shall enforce multi-factor authentication (MFA) for all user logins.	All users	High	Test	Login requires successful second factor; bypass is not possible for standard users.
FR-AUTH-003	Authentication	Password Reset	The system shall provide password reset functionality for internal users.	All users	High	Test	Password reset succeeds via configured channel; reset links/tokens expire; old token cannot be reused.
FR-AUTH-004	Authentication	Account Lockout	The system shall lock user accounts after a configurable number of failed login attempts.	All users	High	Test	After threshold failures, login is blocked until lockout expires or Admin unlocks the account.

FR-AUTH-005	Authentication	Session Timeout	The system shall terminate user sessions after configurable inactivity timeout.	All users	High	Test	After timeout, protected pages require re-authentication; back navigation does not restore authenticated pages.
FR-AUTH-006	Authentication	Logout	The system shall terminate sessions upon user logout.	All users	High	Test	After logout, protected endpoints reject the session token; browser back does not restore access.
FR-AUTH-007	Authentication	Password Storage	The system shall store passwords using secure one-way hashing and shall never store passwords in plaintext.	System	High	Security review	Password hashes are salted/hashed; no plaintext passwords exist in database, logs, or configuration.
FR-AUTHZ-001	Authorization	Protected Module Access	The system shall restrict access to modules and functions based on role permissions.	All authenticated users	High	Test	Unauthorized users cannot access restricted screens or API endpoints; authorized roles can access permitted functions only.
FR-AUTHZ-002	Authorization	Field-Level Security	The system shall support field-level access control for sensitive fields (e.g., bank accounts, national IDs).	Admin; Finance; Operations; M&E	High	Test	Sensitive fields are masked by default; only authorized roles can view unmasked values.
FR-AUTHZ-003	Authorization	Program-Level Data Access	The system shall restrict data access by assigned program(s) where applicable.	Admin; Operations; Finance; M&E	Medium	Test	Users see only records for programs assigned

							to them (if program scoping is enabled).
FR-AUTHZ-004	Authorization	Segregation of Duties	The system shall enforce segregation of duties between disbursement creation (Maker) and approval (Checker/Approver).	Finance	High	Test	A user cannot approve a disbursement they created; workflow blocks self-approval.
FR-AUTHZ-005	Authorization	Read-only Auditor Role	The system shall support a read-only auditor role with access to audit logs and records without edit capability.	Auditor; Admin	Medium	Test	Auditor can view/export reports and logs but cannot modify master or transactional records.
FR-USER-001	User Management	Create User	The system shall allow Admin to create internal user accounts.	Admin	High	Test	New user receives credentials/activation; user can log in when Active.
FR-USER-002	User Management	Update User	The system shall allow Admin to update user profile details (name, email, phone).	Admin	High	Test	Updated user details display correctly; audit log captures the change.
FR-USER-003	User Management	Deactivate User	The system shall allow Admin to deactivate or suspend user accounts.	Admin	High	Test	Deactivated users cannot log in; existing sessions are invalidated or blocked on next request.
FR-ROLE-001	Role Management	Create Role	The system shall allow Admin to create roles.	Admin	High	Test	Role appears in role list and can be assigned to users.
FR-ROLE-002	Role Management	Update Role	The system shall allow Admin to update roles.	Admin	High	Test	Role updates apply to users assigned that role; audit log captures changes.

FR-ROLE-003	Role Management	Deactivate Role	The system shall allow Admin to deactivate roles.	Admin	Medium	Test	Deactivated role cannot be assigned; users previously assigned must be reassigned or lose permissions per policy.
FR-ROLE-004	Role Management	Assign Permissions	The system shall allow Admin to assign granular permissions to roles.	Admin	High	Test	Permission changes immediately affect access; unauthorized screens/APIs are blocked.
FR-USER-004	User Management	Assign Roles to Users	The system shall allow Admin to assign one or more roles to a user.	Admin	High	Test	User access reflects assigned roles; changes are auditable.
FR-USER-005	User Management	Unlock Account	The system shall allow Admin to unlock locked user accounts.	Admin	Medium	Test	Unlocked user can authenticate successfully; action is logged.
FR-PGM-001	Program Management	Create Program	The system shall allow Admin/Operations to create Scholarship programs with independent configurations.	Admin; Operations	High	Test	Program cannot be saved without required fields; appears in program list.
FR-PGM-002	Program Management	Update Program	The system shall allow Admin/Operations to update program details.	Admin; Operations	High	Test	Updated program fields persist and reflect in lists and reports.
FR-PGM-003	Program Management	Archive Program	The system shall allow Admin/Operations to archive/deactivate a program.	Admin; Operations	High	Test	Archived programs cannot be used for new awards/intakes; historical records remain accessible.

FR-PGM-004	Program Management	Configure Academic Periods	The system shall allow Admin/Operations to configure academic periods (year/term/semester) per program.	Admin; Operations	High	Test	Periods are selectable on awards/disbursements; invalid date ranges are rejected.
FR-PGM-005	Program Management	Configure Program Budget	The system shall allow Admin/Operations to configure program budgets and budget ceilings.	Admin; Operations	High	Test	Budget values save; budget utilization updates when awards/disbursements are approved.
FR-PGM-006	Program Management	Budget Guardrail	The system shall prevent approval of awards/disbursements that would exceed program budget ceiling.	Finance; Operations	High	Test	System blocks approval when ceiling would be exceeded and displays a clear validation message.
FR-PGM-007	Program Management	Admin-configurable Settings	The system shall allow authorized administrators to modify program settings without vendor code changes.	Admin	High	Demo & technical documentation	Admin can change configuration via UI; change is effective without redeploying code.
FR-PGM-008	Program Management	Funding Source Allocation	The system shall allow allocation of funding sources/budget lines to programs.	Admin; Operations	Medium	Test	Funding source allocation is selectable during award creation; utilization reports reflect allocations.
FR-PGM-009	Program Management	Configure Award Types	The system shall allow configuration of award types (one-off, recurring, renewable).	Admin; Operations	Medium	Test	Award type settings apply to award lifecycle and disbursement rules.
FR-MD-001	Master Data	Maintain Funding	The system shall allow Admin to create funding source/donor records.	Admin	Medium	Test	Funding source can be created and

		Sources - Create					appears in lists and dropdowns.
FR-MD-002	Master Data	Maintain Funding Sources - Update	The system shall allow Admin to update funding source/donor records.	Admin	Medium	Test	Edits are saved and reflected; audit log captures changes.
FR-MD-003	Master Data	Maintain Funding Sources - View/List	The system shall allow Admin to view and list funding source/donor records.	Admin	Medium	Test	List loads and reflects latest records without duplicates.
FR-MD-004	Master Data	Maintain Funding Sources - Deactivate	The system shall allow Admin to deactivate funding sources/donors.	Admin	Low	Test	Deactivated funding sources cannot be selected for new awards.
FR-SCH-001	Schools	Create School	The system shall allow Admin to create school/institution profiles.	Admin	High	Test	School is created and appears in dropdowns and reports.
FR-SCH-002	Schools	Update School	The system shall allow Admin to update school/institution profiles.	Admin	High	Test	Edits persist; linked records remain valid.
FR-SCH-003	Schools	View/List Schools	The system shall allow authorized users to view and list schools.	Admin; Operations; Finance; M&E	High	Test	Users can view school details per permissions.
FR-SCH-004	Schools	Deactivate School	The system shall allow Admin to deactivate/archieve school profiles.	Admin	Medium	Test	Deactivated schools cannot be selected for new onboarding/awards; historical transactions remain.
FR-BANK-001	Banks	Maintain Banks List	The system shall allow Admin to maintain a list of supported banks.	Admin	Medium	Test	Banks can be created and appear in bank selection fields.

FR-SCHACC-001	School Accounts	Create School Bank Account	The system shall allow Admin to create bank accounts linked to schools with Active/Inactive state.	Admin	High	Test	Account can be created; active accounts are selectable for payments.
FR-SCHACC-002	School Accounts	Update School Bank Account	The system shall allow Admin to update school bank account details.	Admin	High	Test	Updates require approval if configured; changes are audited.
FR-SCHACC-003	School Accounts	Deactivate School Bank Account	The system shall allow Admin to deactivate school bank accounts.	Admin	High	Test	Inactive accounts are not selectable during disbursement.
FR-SCHACC-004	School Accounts	Bank Detail Change Approval	The system shall enforce maker-checker approval for changes to active school bank account details.	Admin; Finance	High	System demo	Change remains pending until approved; audit trail captures maker and approver.
FR-CAT-001	Item Catalog	Create Item	The system shall allow Admin to create disbursement items/categories (e.g., fees, books, uniforms) with Active flag.	Admin	High	Test	Active items appear in disbursement item lists; inactive items do not.
FR-CAT-002	Item Catalog	Update Item	The system shall allow Admin to update disbursement items/categories.	Admin	High	Test	Updates reflect in future selections; historical disbursements retain original item reference.
FR-CAT-003	Item Catalog	Deactivate Item	The system shall allow Admin to deactivate disbursement items/categories.	Admin	Medium	Test	Deactivated items are not selectable for new disbursements.

FR-CAT-004	Item Catalog	View/List Items	The system shall allow authorized users to view and list disbursement items.	Admin; Finance; Operations	Medium	Test	List loads with active/inactive status per permission.
FR-BEN-001	Beneficiaries	Create Beneficiary	The system shall allow authorized users to create beneficiary records.	Operations; Admin	High	Test	Beneficiary can be created with required fields and is visible in registry.
FR-BEN-002	Beneficiaries	Update Beneficiary	The system shall allow authorized users to update beneficiary records.	Operations; Admin	High	Test	Changes are saved; audit log records before/after values.
FR-BEN-003	Beneficiaries	View/List Beneficiaries	The system shall allow authorized users to view and list beneficiary records.	Operations; Finance; M&E; Admin	High	Test	List supports filtering; users see records per access rights.
FR-BEN-004	Beneficiaries	Archive Beneficiary	The system shall allow authorized users to archive/deactivate beneficiary records with reason capture.	Operations; Admin	High	Test	Archived beneficiaries cannot receive new awards/disbursements; history remains accessible.
FR-BEN-005	Beneficiaries	Status Management	The system shall support beneficiary statuses (Imported, Pending Onboarding, Active, Suspended, Closed).	Operations	High	Test	Status transitions follow configured rules; status changes are audited.
FR-BEN-006	Beneficiaries	Prevent Duplicate Beneficiary	The system shall prevent duplicate beneficiaries based on defined uniqueness rules (e.g., National ID + Program + Intake).	Operations; Admin	High	Test	Duplicate attempt is blocked or flagged according to configuration.
FR-IMP-001	Beneficiary Intake	Download CSV Template	The system shall allow authorized users to download standardized CSV templates for beneficiary import.	Operations; Admin	High	System demo	Template downloads successfully and includes required columns and format guidance.

FR-IMP-002	Beneficiary Intake	Upload CSV File	The system shall allow authorized users to upload CSV files for bulk beneficiary import.	Operations	High	System demo	CSV file upload completes; import process starts and shows progress.
FR-IMP-003	Beneficiary Intake	Configurable Field Mapping	The system shall allow configurable mapping between CSV columns and system fields during import.	Operations; Admin	High	System demo	User can map columns; mapping can be saved and reused.
FR-IMP-004	Beneficiary Intake	Validate Required Fields	The system shall validate required fields during import.	Operations	High	System demo	Rows missing required fields fail with a clear reason in error log.
FR-IMP-005	Beneficiary Intake	Validate Data Types	The system shall validate data types/formats during import (dates, numeric fields).	Operations	High	System demo	Invalid formats are rejected with row-level reasons.
FR-IMP-006	Beneficiary Intake	Duplicate Detection on Import	The system shall validate duplicates during import based on configured uniqueness rules.	Operations	High	System demo	Duplicate rows are rejected or flagged; summary shows duplicates.
FR-IMP-007	Beneficiary Intake	Referential Integrity Validation	The system shall validate referential integrity during import (e.g., school, district, program must exist).	Operations	High	System demo	Rows referencing missing master data are rejected with reason.
FR-IMP-008	Beneficiary Intake	Row-level Error Log	The system shall generate a downloadable row-level error log for failed import records.	Operations	High	System demo	Error log includes row number and failure reason; file is downloadable.
FR-IMP-009	Beneficiary Intake	Import Summary Report	The system shall generate an import summary (total, accepted, rejected).	Operations	High	System demo	Summary counts match imported dataset; report is saved in import history.
FR-IMP-010	Beneficiary Intake	Partial Success Import	The system shall import valid rows without being blocked by rejected rows.	Operations	High	System demo	Accepted records appear in registry

							even when some rows fail.
FR-ONB-001	Onboarding	Convert Imported to Beneficiary Profile	The system shall allow authorized users to convert an imported record into an onboarded beneficiary profile.	Operations	High	System demo	Converted record becomes available for award creation; onboarding action is audited.
FR-ONB-002	Onboarding	Mandatory Onboarding Fields	The system shall enforce completion of mandatory onboarding fields before activation.	Operations	High	System demo	Activation is blocked until mandatory fields are complete.
FR-ONB-003	Onboarding	Onboarding Document Upload	The system shall allow uploading onboarding documents (e.g., admission letter, ID) linked to beneficiary.	Operations	High	System demo	Documents are stored and visible in beneficiary document list.
FR-ONB-004	Onboarding	Internal Validation Confirmation	The system shall allow internal validation confirmation prior to onboarding approval.	Operations	High	System demo	Record cannot be approved for onboarding without validation confirmation; actions are audited.
FR-ONB-005	Onboarding	Exception Queue	The system shall allow users to flag beneficiary records as exceptions with reason capture.	Operations	Medium	System demo	Exception appears in exception queue; reason is recorded and auditable.
FR-ONB-006	Onboarding	Resolve Exceptions	The system shall allow authorized users to resolve flagged exceptions.	Operations	Medium	System demo	Resolved exception is removed from queue; resolution action is audited.
FR-DOC-001	Document Management	Upload Beneficiary Document	The system shall allow authorized users to upload	Operations; Admin	High	Test	Uploaded document is linked to beneficiary

			documents linked to beneficiary records.				and can be retrieved by authorized roles.
FR-DOC-002	Document Management	Upload Disbursement Evidence	The system shall allow authorized users to upload payment evidence linked to disbursement records.	Finance	High	Test	Evidence is linked to the disbursement and visible in disbursement details.
FR-DOC-003	Document Management	Virus Scan on Upload	The system shall virus-scan files upon upload.	System	High	System demo	Infected files are rejected; event is logged.
FR-DOC-004	Document Management	File Size Limits	The system shall enforce configurable file size limits for uploads.	System; Admin	Medium	Test	Files above limit are rejected with a user-friendly message.
FR-DOC-005	Document Management	Document Metadata	The system shall store document metadata (type, owner record, uploaded by, upload date).	System	High	Test	Metadata is visible to authorized users and included in audit trails.
FR-DOC-006	Document Management	View Document Metadata	The system shall allow authorized users to view document metadata.	Operations; Finance; Admin; Auditor	High	Test	Metadata displays correctly and respects access permissions.
FR-DOC-007	Document Management	Download Document	The system shall allow authorized users to download documents.	Authorized users	High	Test	Download succeeds for authorized users; unauthorized users are blocked.
FR-DOC-008	Document Management	Log Document Downloads	The system shall log all document download activities.	System	High	Test	Log includes user, timestamp, document, and record reference.
FR-DOC-009	Document Management	Document Status - Pending	The system shall allow authorized users to set document status to Pending.	Operations	Medium	Test	Status change is recorded and visible.

FR-DOC-010	Document Management	Document Status - Verified	The system shall allow authorized users to set document status to Verified.	Operations	High	Test	Verified documents can satisfy mandatory document rules.
FR-DOC-011	Document Management	Document Status - Rejected	The system shall allow authorized users to set document status to Rejected.	Operations	High	Test	Rejected documents are marked clearly and require re-upload where applicable.
FR-DOC-012	Document Management	Rejection Reason Required	The system shall require capture of rejection reason when a document is marked Rejected.	Operations	High	Test	Rejection cannot be saved without a reason; reason is visible to authorized users.
FR-DOC-013	Document Management	Document Versioning	The system shall support uploading a new version of an existing document.	Operations; Finance	Medium	Test	New version is linked; previous versions remain accessible for audit.
FR-DOC-014	Document Management	Document Version History	The system shall display document version history.	Authorized users	Medium	Test	Version history shows all versions with upload dates and users.
FR-DOC-015	Document Management	Document Expiry Date	The system shall allow setting expiry dates for documents where applicable.	Operations; Admin	Low	Test	Expired documents are flagged.
FR-DOC-016	Document Management	Flag Expired Documents	The system shall automatically flag expired documents.	System	Low	Test	Expired status appears and can trigger re-upload workflow if enabled.
FR-DOC-017	Document Management	Prevent Deletion After Approval	The system shall prevent deletion of documents linked to approved financial transactions.	Admin; Finance	High	Test	Delete option is not available or is blocked for approved transactions; attempt is audited.

FR-DOC-018	Document Management	Controlled Document Deletion	The system shall allow controlled deletion of documents prior to transaction approval based on permissions.	Admin	Medium	Test	Only authorized role can delete; deletion is logged with reason.
FR-AWARD-001	Award Management	Create Award	The system shall allow authorized users to create Scholarship awards linked to approved beneficiaries and programs.	Operations; Finance	High	Test	Award creation requires active beneficiary; award appears in beneficiary profile.
FR-AWARD-002	Award Management	View/List Awards	The system shall allow authorized users to view and list awards.	Operations; Finance; Admin	High	Test	Award list supports filters by program, period, status.
FR-AWARD-003	Award Management	Update Award Before First Disbursement	The system shall allow updates to award details prior to first disbursement, subject to permissions.	Operations; Finance	Medium	Test	Award can be updated; audit log captures changes.
FR-AWARD-004	Award Management	Protect Award Amount After Payment	The system shall restrict award amount changes after first disbursement unless approved by an authorized role.	Finance; Admin	High	System demo	System blocks changes or requires approval with justification; action is auditable.
FR-AWARD-005	Award Management	Activate Award	The system shall allow authorized users to activate an award.	Operations	High	Test	Inactive awards cannot be used for disbursement; activation enables disbursement.
FR-AWARD-006	Award Management	Suspend Award	The system shall allow authorized users to suspend an award with mandatory reason capture.	Operations	High	Test	Suspended award cannot receive new disbursements; reason is stored.

FR-AWARD-007	Award Management	Reinstate Award	The system shall allow authorized users to reinstate a suspended award.	Operations	Medium	Test	Reinstated award becomes eligible for disbursement.
FR-AWARD-008	Award Management	Close Award	The system shall allow authorized users to close/terminate an award with mandatory reason capture.	Operations	High	Test	Closed awards cannot accept new disbursements; closure reason is recorded.
FR-AWARD-009	Award Management	Generate Award Letter	The system shall generate award confirmation letters using configurable templates.	Operations	Low	System demo	Generated letter contains correct variable fields and is stored in beneficiary record.
FR-AWARD-010	Award Management	Renew Award	The system shall allow authorized users to renew awards for a new period where renewal applies.	Operations	Low	System demo	Renewal creates a new period record linked to original award; history preserved.
FR-FIN-001	Disbursements	Create Disbursement Request	The system shall allow Finance to create disbursement requests linked to active awards.	Finance	High	Test	Disbursement cannot be created for inactive beneficiary/award; valid requests are saved.
FR-FIN-002	Disbursements	Specify Payee Model	The system shall allow Finance to specify payee type (school/guardian/vendor) for a disbursement.	Finance	High	Test	Payee selection is validated; payee details auto-populate from master data.
FR-FIN-003	Disbursements	Specify Disbursement Category	The system shall allow Finance to select disbursement item/category from approved catalog.	Finance	High	Test	Only active catalog items are selectable.

FR-FIN-004	Disbursements	Budget Balance Check	The system shall prevent disbursement amounts from exceeding remaining award balance.	Finance	High	Test	System blocks amounts above remaining balance; message shows available balance.
FR-FIN-005	Disbursements	Program Budget Check	The system shall prevent disbursement approvals that exceed program budget availability.	Finance	High	Test	Approval is blocked when budget insufficient.
FR-FIN-006	Disbursements	Multi-level Approval	The system shall support multi-level approval workflow for disbursements (Maker-Checker-Approver).	Finance	High	System demo	Disbursement cannot be marked Approved/Paid without required approvals; approval trail captured.
FR-FIN-007	Disbursements	Approval Trail	The system shall record approval timestamps and approving user details for disbursements.	Finance; Auditor	High	Test	Disbursement shows full approval history.
FR-FIN-008	Disbursements	Prevent Self-Approval	The system shall prevent a user from approving a disbursement they created.	Finance	High	Test	System blocks self-approval attempt.
FR-FIN-009	Disbursements	Disbursement Status Tracking	The system shall support disbursement statuses (Requested, Approved, Paid, Failed, Reconciled).	Finance	High	Test	Status transitions are controlled; statuses display consistently in lists and reports.
FR-FIN-010	Disbursements	Failure Reason Capture	The system shall require a reason when marking a disbursement as Failed.	Finance	High	Test	Failed status cannot be saved without reason; reason appears in audit trail.
FR-FIN-011	Disbursements	Attach Payment Evidence	The system shall allow attachment of payment evidence	Finance	High	Test	Evidence is attached and required before

			(voucher/receipt/bank confirmation) to disbursements.				status can be set to Paid (if configured).
FR-FIN-012	Disbursements	Evidence Required for Paid	The system shall prevent marking a disbursement as Paid unless payment evidence is attached (configurable).	Finance	High	System demo	Attempt to mark Paid without evidence is blocked with user-friendly message.
FR-FIN-013	Disbursements	Duplicate Payment Prevention	The system shall prevent duplicate disbursements for the same beneficiary, category, and academic period.	Finance	High	Test	Duplicate attempt is blocked or requires authorized override with justification if enabled.
FR-FIN-014	Disbursements	Duplicate Override with Justification	The system shall allow authorized override of duplicate disbursement prevention with mandatory justification.	Finance; Admin	Low	System demo	Override requires justification; override action is audited.
FR-FIN-015	Disbursements	Batch Disbursement	The system shall allow Finance to create a batch of disbursements for multiple beneficiaries and/or items.	Finance	Medium	System demo	Batch creates individual disbursement records; batch summary totals match line items.
FR-FIN-016	Disbursements	Multiple Disbursements Allowed	The system shall allow multiple disbursements for the same award without overwriting prior records.	Finance	High	Test	Multiple disbursements appear in history and reports distinctly.
FR-FIN-017	Disbursements	Bank Detail Lock During Processing	The system shall retrieve payee bank details from approved master data and prevent editing during disbursement processing.	Finance	High	Test	Bank details cannot be edited on disbursement screen; changes must occur through master data workflow.

FR-FIN-018	Reversals & Adjustments	Record Returned Funds	The system shall allow Finance to record returned funds linked to a disbursement.	Finance	Medium	System demo	Returned funds entry reduces paid totals and is auditable.
FR-FIN-019	Reversals & Adjustments	Record Reversal	The system shall allow Finance to record a disbursement reversal.	Finance	Medium	System demo	Reversal updates balances correctly; reversal reason is captured.
FR-FIN-020	Reconciliation	Mark Reconciled	The system shall allow Finance to mark disbursements as reconciled.	Finance	High	System demo	Reconciled status is saved and included in reconciliation reports.
FR-FIN-021	Reconciliation	Prevent Changes After Reconciliation	The system shall prevent modification of reconciled disbursement records.	Finance; Admin	High	Test	Edit actions are blocked once reconciled; attempt is logged.
FR-FIN-022	Reconciliation	Reconciliation Reports	The system shall generate disbursement reconciliation reports.	Finance; Auditor	High	System demo	Report totals match reconciled transaction dataset and filters applied.
FR-ME-001	Monitoring & Evaluation	Record Performance	The system shall allow M&E users to record academic performance per beneficiary per academic period.	M&E	High	Test	Performance record saves and is visible in beneficiary history.
FR-ME-002	Monitoring & Evaluation	Update Performance	The system shall allow authorized users to update academic performance records.	M&E	Medium	Test	Edits are tracked in audit log.
FR-ME-003	Monitoring & Evaluation	Record Attendance	The system shall allow M&E users to record attendance status per beneficiary per period.	M&E	Medium	Test	Attendance record saves; period is required.

FR-ME-004	Monitoring & Evaluation	Record Progression	The system shall allow M&E users to record progression status (Promoted/Repeated/Completed/Dropped).	M&E	Medium	Test	Progression changes are recorded and reportable.
FR-ME-005	Risk Management	Auto Flag Low Performance	The system shall automatically flag beneficiaries whose performance falls below configured thresholds (optional).	M&E	Low	System demo	System flags beneficiary and places in at-risk list when criteria met.
FR-ME-006	Risk Management	Manual At-Risk Flag	The system shall allow M&E users to manually flag a beneficiary as at-risk with reason capture.	M&E	High	Test	At-risk flag requires reason; appears in at-risk register.
FR-ME-007	Case Management	Log Intervention	The system shall allow M&E users to log interventions/actions linked to a beneficiary.	M&E	High	Test	Intervention record captures action, owner, due date, and status.
FR-ME-008	Case Management	Assign Intervention Owner	The system shall allow assignment of responsibility for interventions.	M&E	Medium	Test	Assigned owner is stored and can be filtered in intervention views.
FR-ME-009	Case Management	Update Intervention Status	The system shall allow updating intervention status (Open/In Progress/Closed).	M&E	Medium	Test	Status changes are captured and auditable.
FR-ME-010	Case Management	Prevent Intervention Deletion	The system shall prevent deletion of intervention records.	System	High	Test	Delete action not available; attempts are blocked and logged.
FR-ME-011	Monitoring Visits	Record Site Visit	The system shall allow recording monitoring visits linked to beneficiaries or schools.	M&E	Low	System demo	Visit record includes date, location, and linked entity.

FR-ME-012	Monitoring Visits	Upload Visit Report	The system shall allow uploading visit reports and attachments.	M&E	Low	System demo	Attachments are linked to visit record; access controlled.
FR-ME-013	Outcomes	Record Completion	The system shall allow recording completion/graduation outcomes for beneficiaries.	M&E	Medium	Test	Outcome captured and appears in outcomes reporting.
FR-ME-014	Outcomes	Record Exit Reason	The system shall require exit reason capture for discontinued beneficiaries.	M&E	Medium	Test	Exit reason is mandatory when setting status to Dropped/Closed.
FR-ME-015	Outcomes	Outcome Metrics	The system shall calculate and report program outcome metrics (completion, dropout, progression rates).	M&E; Management	Medium	System demo	Metrics match underlying dataset and applied filters.
FR-DASH-001	Dashboard	Management Dashboard	The system shall provide a dashboard showing key Scholarship KPIs (active beneficiaries, budget utilization, disbursements).	Management; Admin; Finance; Operations	Medium	System demo	KPIs load and reflect current data; role-based visibility applies.
FR-DASH-002	Dashboard	Recent Disbursements Widget	The system shall display a recent disbursements widget based on disbursement transactions.	Finance; Management	Medium	System demo	New disbursement appears immediately after capture/approval per refresh.
FR-DASH-003	Dashboard	At-risk Summary	The system shall display at-risk beneficiary counts (optional).	M&E; Management	Low	System demo	At-risk count matches at-risk register.
FR-REP-001	Reporting	Beneficiary Register Report	The system shall provide a beneficiary register report.	Operations; Management; Auditor	High	Test	Report totals match beneficiary registry filtered dataset.
FR-REP-002	Reporting	Awards Report	The system shall provide an awards report by program and period.	Operations; Finance	High	Test	Awards totals reconcile to award master records.

FR-REP-003	Reporting	Disbursements Report	The system shall provide disbursement reports by program, period, payee, and category.	Finance; Auditor	High	Test	Report totals reconcile to disbursement transactions.
FR-REP-004	Reporting	Budget Utilization Report	The system shall provide program budget utilization reports.	Management; Finance; Operations	High	Test	Utilization equals approved awards/disbursements per configuration.
FR-REP-005	Reporting	M&E Outcomes Report	The system shall provide M&E outcome reports (completion, dropout, progression).	M&E; Management	Medium	System demo	Outcome measures match stored outcomes and period filters.
FR-REP-006	Reporting	Report Filters	The system shall allow filtering reports by program, academic period, district, school, and status.	Authorized users	High	System demo	Filters update results correctly and are reflected in exports.
FR-REP-007	Reporting	Dynamic Report Builder	The system shall provide a dynamic report builder with drag-and-drop field selection for authorized users.	Admin; Finance; M&E; Operations	High	System demo	User can drag fields into layout and generate a report output.
FR-REP-008	Reporting	Save Report Template	The system shall allow users to save dynamic report definitions as reusable templates.	Admin; Finance; M&E; Operations	Medium	System demo	Saved template appears in templates list and can be re-run.
FR-REP-009	Reporting	Edit Report Template	The system shall allow users to edit saved report templates.	Admin; Finance; M&E; Operations	Low	System demo	Edits persist; template versioning/audit is recorded if enabled.
FR-REP-010	Reporting	Delete Report Template	The system shall allow users to delete saved report templates based on permissions.	Admin	Low	System demo	Deleted template is removed from list; deletion is audited.

FR-REP-011	Reporting	Export Reports - CSV	The system shall allow exporting report outputs to CSV.	Authorized users	High	Test	Exported CSV matches filtered dataset; export event is logged.
FR-REP-012	Reporting	Export Reports - Excel	The system shall allow exporting report outputs to Excel.	Authorized users	High	Test	Exported Excel matches filtered dataset; export event is logged.
FR-REP-013	Reporting	Export Reports - PDF	The system shall allow exporting report outputs to PDF (where supported).	Authorized users	Medium	System demo	Exported PDF contains correct headings and filters; export event is logged.
FR-REP-014	Reporting	Export Logging	The system shall log all report and data exports (user, date/time, report name, format).	System; Auditor	High	Test	Export log is viewable by auditor/admin and includes necessary metadata.
FR-REP-015	Reporting	Scheduling Reports (Optional)	The system shall allow scheduling of report generation and delivery to internal users (optional).	Admin	Low	System demo	Scheduled report runs at configured time and delivers to configured recipients; failures are logged.
FR-NOTIF-001	Notifications	Email Notifications	The system shall send automated email notifications for configured internal events.	System; Admin	Medium	System demo	Email is sent on trigger; delivery status is logged.
FR-NOTIF-003	Notifications	Notification Templates	The system shall allow Admin to create and edit notification templates.	Admin	Medium	System demo	Template updates are used for subsequent notifications; changes are auditable.
FR-NOTIF-004	Notifications	Notification Trigger Control	The system shall allow Admin to activate/deactivate notification triggers.	Admin	Medium	System demo	Disabled trigger does not send notifications; action is audited.

FR-NOTIF-005	Notifications	Notification Log	The system shall maintain a log of sent notifications including recipient, channel, timestamp, and status.	Admin; Auditor	Medium	Test	Notification log reflects sent status and failures with reason (where available).
FR-API-001	Integration	Secure API Authentication	The system shall require secure authentication and authorization for all API endpoints.	System	High	Security review	Unauthenticated requests are rejected; role-based API access is enforced.
FR-API-002	Integration	API Access Logging	The system shall log all API access activities.	System; Auditor	Medium	Test	Logs contain endpoint, user/client, timestamp, status code, and correlation ID (where supported).
FR-API-003	Integration	Disbursement Data API (Optional)	The system shall provide secure API endpoints to retrieve approved disbursement data (optional).	System; Finance	Low	Technical documentation & demo	API returns only authorized data; data matches UI/reporting dataset.
FR-EMAIL-001	Email Settings	Maintain SMTP Settings	The system shall allow Admin to configure SMTP settings.	Admin	Medium	Test	Valid SMTP config enables email delivery; config changes are audited.
FR-EMAIL-002	Email Settings	Encrypt SMTP Credentials	The system shall store SMTP credentials encrypted at rest.	System	High	Security review	Credentials are not visible in UI or logs; stored encrypted in database or secret store.
FR-AUD-001	Audit & Compliance	Audit Log - Capture Actions	The system shall capture audit logs for key actions (create, update, delete/archive, approve, export).	System	High	Test	Audit entries include actor, timestamp, action, entity, and before/after values where applicable.

FR-AUD-002	Audit & Compliance	Audit Log - Tamper Protection	Audit logs shall be protected from unauthorized modification or deletion.	Admin; System	High	Security review	Users cannot edit audit logs; attempts are blocked and logged.
FR-AUD-003	Audit & Compliance	Audit Log Retention	The system shall retain audit logs for a configurable period per organizational policy.	Admin; System	Medium	Test	Retention period is configurable; logs are retrievable for the full retention window.
FR-AUD-004	Audit & Compliance	Audit Log Admin Restriction	Access to audit log configuration and management shall be restricted to authorized administrative roles only.	Admin	Medium	Test	Only designated admin roles can access audit configuration; access attempts are logged.
FR-DG-001	Data Governance	Consistent Validation Rules	The system shall enforce consistent validation rules across manual entry, CSV import, and API interfaces.	System	High	Test	Same invalid data is rejected across UI, import, and API with consistent reasons.
FR-DG-002	Data Governance	Prevent Deletion of Financial Records	The system shall prevent deletion of records linked to financial transactions.	System	High	Test	Delete action is blocked; record can be archived if permitted; attempt is auditable.
FR-DG-003	Data Governance	Exception Queue for Data Quality	The system shall support exception queues for resolution of flagged data inconsistencies.	Operations; Admin	Medium	System demo	Exceptions can be filtered, assigned, and resolved with auditable notes.
FR-DG-004	Data Governance	Data Export in Open Formats	The system shall allow export of core datasets in open formats (CSV/Excel) subject to permissions.	Admin; Auditor; Authorized users	Medium	System demo	Exports succeed for authorized users; exports are logged and include applied filters.

FR-OPS-001	Operability	Alert on Logging Failure	The system shall generate alerts in the event of audit logging failure.	System; Admin	Medium	Technical review	Alert is triggered when logging fails; event is recorded for support.
FR-OPS-002	Operability	Alert on Backup Failure	The system shall generate alerts when backups fail.	System; Admin	Medium	Technical review	Backup failure alert is generated; failure is logged.
FR-OPS-003	Operability	Time Synchronization	The system shall synchronize system time with an approved time source to ensure accurate audit timestamps.	System	Low	Technical review	Time sync is configured; audit timestamps are consistent across components.

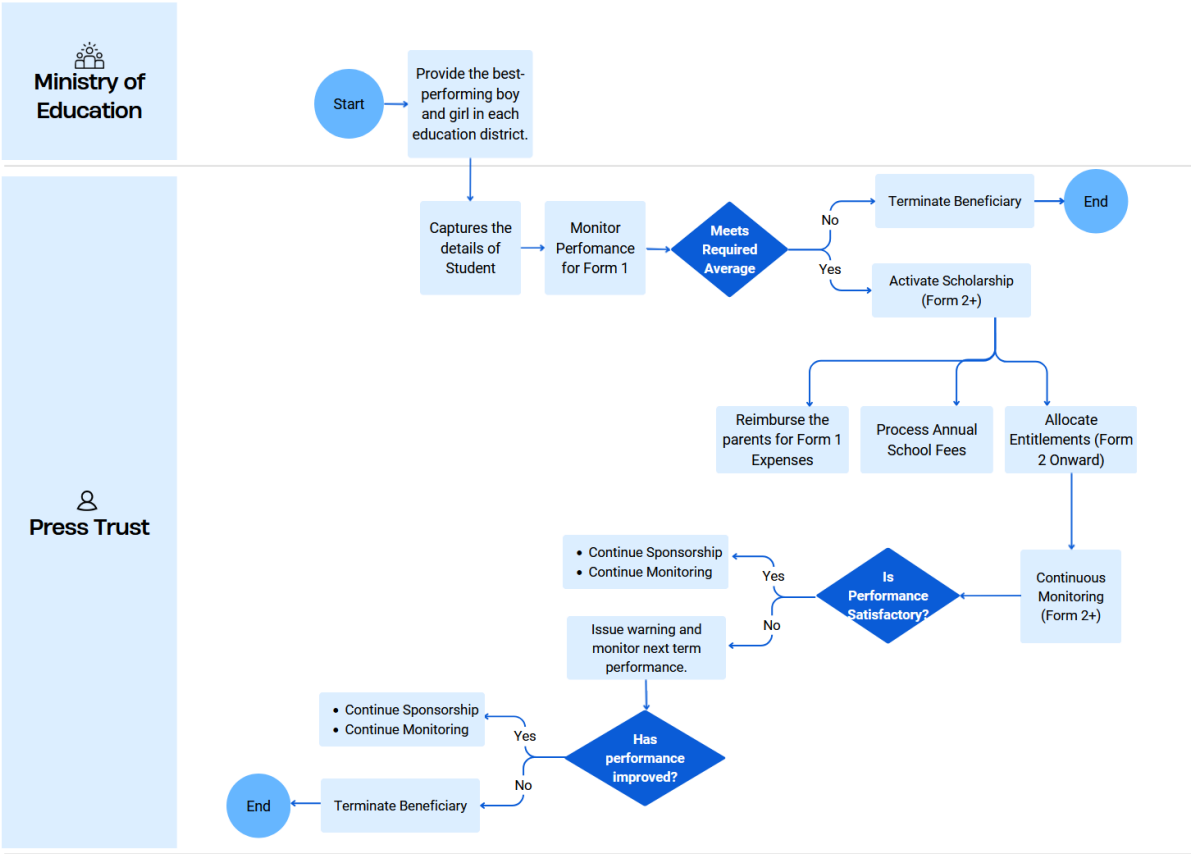
17.2 Appendix B: Requirements Traceability (High-Level)

The table below presents the high-level functional requirements referenced in Section 3. These requirements are not exhaustive and may be refined or supplemented during system demonstrations and implementation discussions. During implementation, the vendor shall map each listed requirement to the relevant design components, test cases, and release versions. Vendors may propose additional requirements or enhancements; however, all requirements specified herein shall remain mandatory and shall not be removed or substituted.

Req ID	Description	Priority	Verification
FR-01	Create and manage multiple Scholarship programs with independent configurations.	Must	System demo
FR-02	Configure academic periods, funding cycles, and program budgets per Scholarship program.	Must	System demo
FR-03	Configure award types (one-off, recurring, renewable) including duration and lifecycle rules.	Must	System demo
FR-04	Allow program configuration changes without vendor code changes (admin-configurable).	Must	Demo & technical documentation
FR-05	Role-based access control (RBAC) for Operations, Finance, M&E, and Admin users.	Must	System demo
FR-06	Restrict data visibility based on role permissions and assigned responsibilities.	Must	System demo
FR-07	Enforce segregation of duties between disbursement creation and approval.	Must	System demo
FR-08	Maintain user activity audit logs for all key system actions.	Must	System demo
FR-09	Bulk import government-approved beneficiary lists via CSV upload.	Must	System demo
FR-10	Validate imported records for required fields, duplicates, data types, and referential integrity.	Must	System demo
FR-11	Generate row-level error logs and import summary reports.	Must	System demo
FR-12	Maintain beneficiary master records including demographics and identifiers.	Must	System demo
FR-13	Capture and manage school/institution profiles.	Must	System demo
FR-14	Capture and securely store banking details with masking and access controls.	Must	Demo & security review
FR-15	Upload and store supporting documents linked to beneficiaries and disbursements.	Must	System demo
FR-16	Support document verification workflow (Pending/Verified/Rejected) with audit trail.	Must	System demo
FR-17	Restrict document access by role and log document downloads.	Must	System demo
FR-18	Support document versioning and expiry tracking where applicable.	Must	System demo
FR-19	Create Scholarship awards linked to approved beneficiaries.	Must	System demo
FR-20	Track award lifecycle from activation to closure with mandatory reason capture.	Must	System demo

FR-21	Generate award letters using configurable templates.	Must	System demo
FR-22	Support award renewals based on defined program rules.	Must	System demo
FR-23	Record approved award amounts and disbursement schedules.	Must	System demo
FR-24	Track support/expenditure categories (fees, books, uniforms, etc.).	Must	System demo
FR-25	Multi-level approval workflow for disbursements (Maker-Checker-Approver).	Must	System demo
FR-26	Track disbursement status (Requested/Approved/Paid/Reconciled/Failed).	Must	System demo
FR-27	Attach payment evidence to each disbursement record.	Must	System demo
FR-28	Support payments to schools, guardians, or vendors as configured.	Must	System demo
FR-29	Prevent duplicate disbursements for same beneficiary and academic period.	Must	System demo
FR-30	Maintain full audit trail of financial transactions and approvals.	Must	System demo
FR-31	Maintain academic performance indicators over time (results, attendance, progression).	Must	System demo
FR-32	Flag at-risk beneficiaries and log interventions with follow-up reminders.	Must	System demo
FR-33	Capture monitoring/site visit reports including findings and attachments.	Must	System demo
FR-34	Track program outcomes and impact metrics (completion, dropout, graduation, KPIs).	Must	System demo
FR-35	Provide predefined operational and financial reports.	Must	System demo
FR-36	Provide dynamic ad-hoc reporting with drag-and-drop field selection.	Must	System demo
FR-37	Allow filtering by program, period, school, district, and status.	Must	System demo
FR-38	Export reports to Excel, PDF, and CSV including applied filters.	Must	System demo
FR-39	Provide dashboards with KPIs and trend views for management.	Must	System demo
FR-40	Provide automated notifications (email) for key internal system events.	Must	System demo
FR-41	Provide secure API integration capabilities where required.	Must	Technical documentation & demo
FR-42	Support configurable SMTP gateway settings.	Must	System demo
FR-43	Enforce consistent validation rules across manual entry, CSV import, and API interfaces.	Must	System demo
FR-44	Provide duplicate detection mechanisms for beneficiaries and payees.	Must	System demo
FR-45	Prevent deletion of records linked to financial transactions.	Must	System demo

18. Process Flow Diagram



Business Process flow Fig 1

19. Minimum Vendor Team Composition and Qualification Requirements

1. Project Manager

Minimum Qualifications

- Bachelor's Degree in Information Technology, Computer Science, Engineering, or Business.
- Professional certification (mandatory advantage):
 - PMP / PRINCE2 / Agile Certification.

Minimum Experience

- At least 8 years' experience in IT project management.
- At least 3 successfully delivered enterprise system implementations.
- Proven experience managing:
 - Multi-stakeholder environments.
 - Projects involving financial systems or workflow automation.

Key Competencies

- Risk management and mitigation.
- Stakeholder engagement (Board, management, users).
- Project governance and reporting.

2. Solution Architect / Technical Lead

Minimum Qualifications

- Bachelor's Degree in Computer Science, Software Engineering, or related field
- Relevant certifications (advantage):
 - Cloud (AWS/Azure).
 - TOGAF or equivalent architecture certification.

Minimum Experience

- At least 8 years' experience in system architecture and software development.
- At least 3 enterprise system designs (web-based platforms).
- Proven experience in:
 - Secure system design (RBAC, encryption, MFA).
 - API-based integrations.
 - Scalable systems.

Key Competencies

- End-to-end system design.
- Integration architecture.
- Security-by-design principles.

3. Software Developer - Backend

Minimum Qualifications

- Degree/Diploma in Computer Science, Software Engineering, or equivalent.

Minimum Experience

- At least 5 years' experience in backend development.
- Proven experience in:
 - Workflow-based systems.
 - API development (RESTful services).

- Database interaction.

Key Competencies

- Business logic implementation.
- Secure coding practices.
- Performance optimization.

4. Software Developer - Frontend

Minimum Qualifications

- Degree/Diploma in IT or Software Development.

Minimum Experience

- At least 5 years' experience in frontend/web development.
- Proven experience developing:
 - Responsive, mobile-friendly applications.
 - Low-bandwidth optimized interfaces.

Key Competencies

- User interface design.
- Usability and accessibility.
- Cross-browser compatibility.

5. Business Analyst / Systems Analyst

Minimum Qualifications

- Degree in Information Systems, Business IT, or related field.

Minimum Experience

- At least 5 years' experience in business/system analysis.
- Proven experience in:
 - Process mapping (DFDs, workflows).
 - Requirements gathering and validation.
 - Translating business processes into system specifications.

Key Competencies

- Workflow design (critical for scholarship lifecycle).
- Stakeholder requirement management.
- Documentation and validation.

6. Database Administrator (DBA)

Minimum Qualifications

- Degree in Computer Science, IT, or Database Management.

Minimum Experience

- At least 5 years' experience in database administration
- Proven experience in:
 - RDBMS (MySQL, PostgreSQL, SQL Server, Oracle).
 - Data migration and reconciliation.
 - Backup and disaster recovery.

Key Competencies

- Data integrity and security.
- Performance tuning.
- Migration of legacy data (Excel-based systems).

7. Quality Assurance (QA) / Testing Specialist

Minimum Qualifications

- Degree/Diploma in IT, Software Testing, or related field.

Minimum Experience

- At least 5 years' experience in software testing.
- Proven experience in:
 - Functional testing.
 - User Acceptance Testing (UAT).
 - Test plan and test case development.

Key Competencies

- Defect management.
- Test automation (advantage).
- Ensuring system reliability and stability.

20. Ownership, Intellectual Property & Sustainability

The solution may be delivered as either a custom-developed system or a commercial off-the-shelf (COTS/SaaS) product. The following conditions shall apply:

1. Data Ownership

Press Trust shall retain full and exclusive ownership of all data stored, processed, or generated within the system at all times.

The vendor shall provide the ability to export all core datasets (beneficiaries, awards, disbursements, monitoring records, audit logs, and reference data) in open, standard formats (e.g., CSV, Excel) without restriction.

2. Custom-Developed Components

Where the system or any part thereof is custom-developed specifically for Press Trust:

- Press Trust shall receive full ownership rights to the custom-developed components.
- The vendor shall hand over complete and up-to-date source code.
- The handover shall include database schemas, configuration files, deployment scripts, and technical documentation.
- The source code shall be structured, documented, and free from hard-coded credentials.
- Press Trust shall retain unrestricted rights to use, modify, and further develop the custom-developed components.

3. Commercial Off-the-Shelf (COTS/SaaS) Components

Where the solution includes commercial off-the-shelf components:

- The vendor shall provide Press Trust with appropriate licensing or subscription rights.
- Press Trust shall retain full ownership of its data.
- The system shall not impose vendor lock-in.
- The vendor shall provide full administrative access required for configuration and management.
- The vendor shall provide complete documentation and API documentation where applicable.
- In the event of contract termination, the vendor shall provide full data extraction support within an agreed transition period.

4. Business Continuity & Exit Support

Upon termination or expiration of the contract:

- The vendor shall provide data export in open formats.
- The vendor shall provide reasonable transition assistance.
- The vendor shall ensure no proprietary mechanisms prevent data migration.

21. Recommended Implementation Timeline

Phase	Timeline
Requirements & System Design	2–3 weeks
Development & Configuration	5–7 weeks
Integration, Testing & Deployment	3–4 weeks
Post-Implementation Support	12 months after Go-Live

Total implementation period: approximately 3–4 months, followed by 12 months support.

22. Milestone-Based Payment Plan

The implementation of the Scholarship Management System shall follow a milestone-based payment structure tied to the successful completion and acceptance of key deliverables. Payments shall be made upon certification by the client that each milestone has been satisfactorily achieved.

Milestone	Deliverables	Payment
Requirements & System Design	Approved System Requirement Specification and system architecture	25%
System Development	Completion of system modules and prototype demonstration	30%
Integration & Deployment	Successful integration, testing, deployment, and training	30%
Post-Implementation Support	Provision of technical support and maintenance for 12 months	15%

23. Document Approval

The requirements listed above accurately reflect the agreed discussions and the intended operation of the proposed system. I hereby approve the Software Requirements Specification (SRS).

NAME	ROLE	TITLE	SIGNATURE	DATE
Gibson Ngalamila	Project Sponsor\Owner	Executive Secretary		
Madalitso Kazembe	Process Owner	Head of Operations		
Johannes Kumpata	Stakeholder	Operations Manager		
Chimwemwe Songwe	Stakeholder	Projects Officer		
Sherry Dukes	Stakeholder	Executive Assistant		
Gift Mtonga	Stakeholder	Intern Quantity Surveyor		
Barry Makumba	ICT Consultant	Lead IT Systems Consultant		